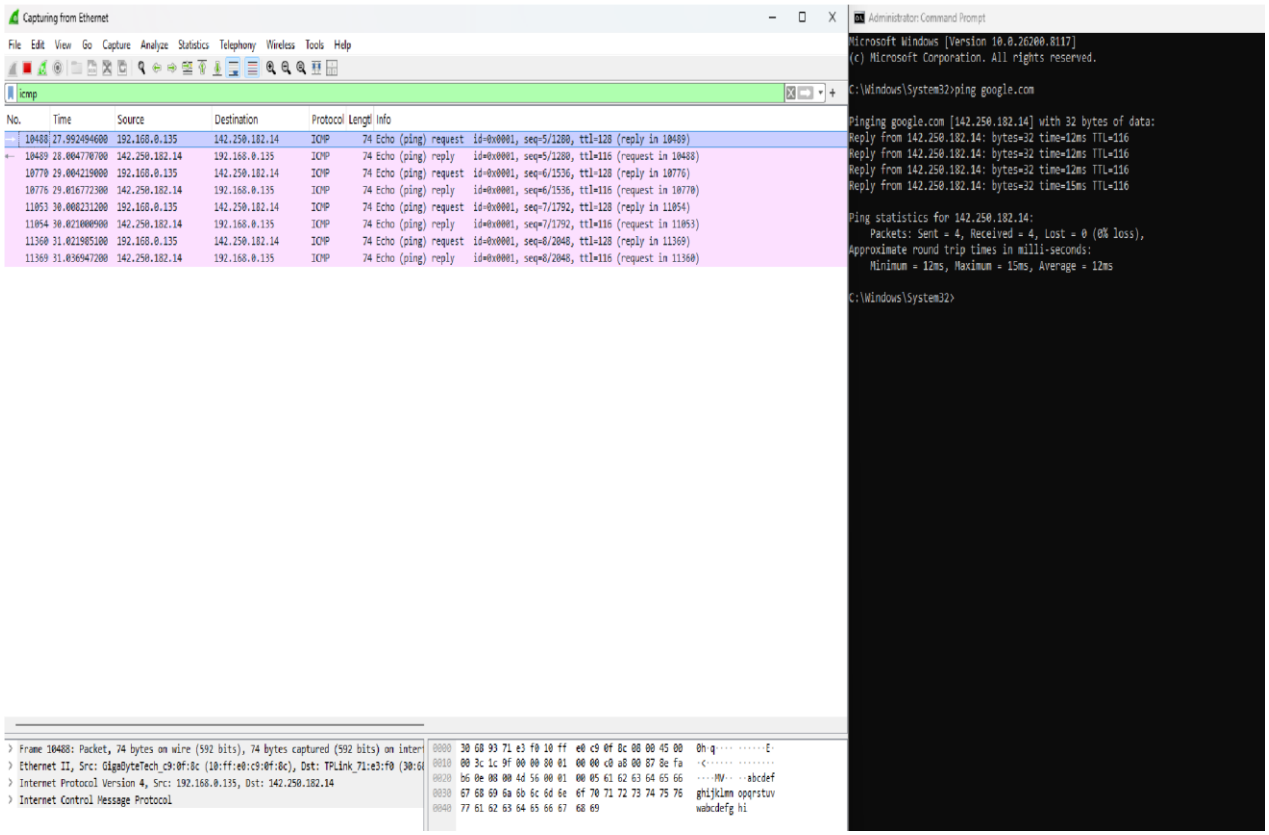
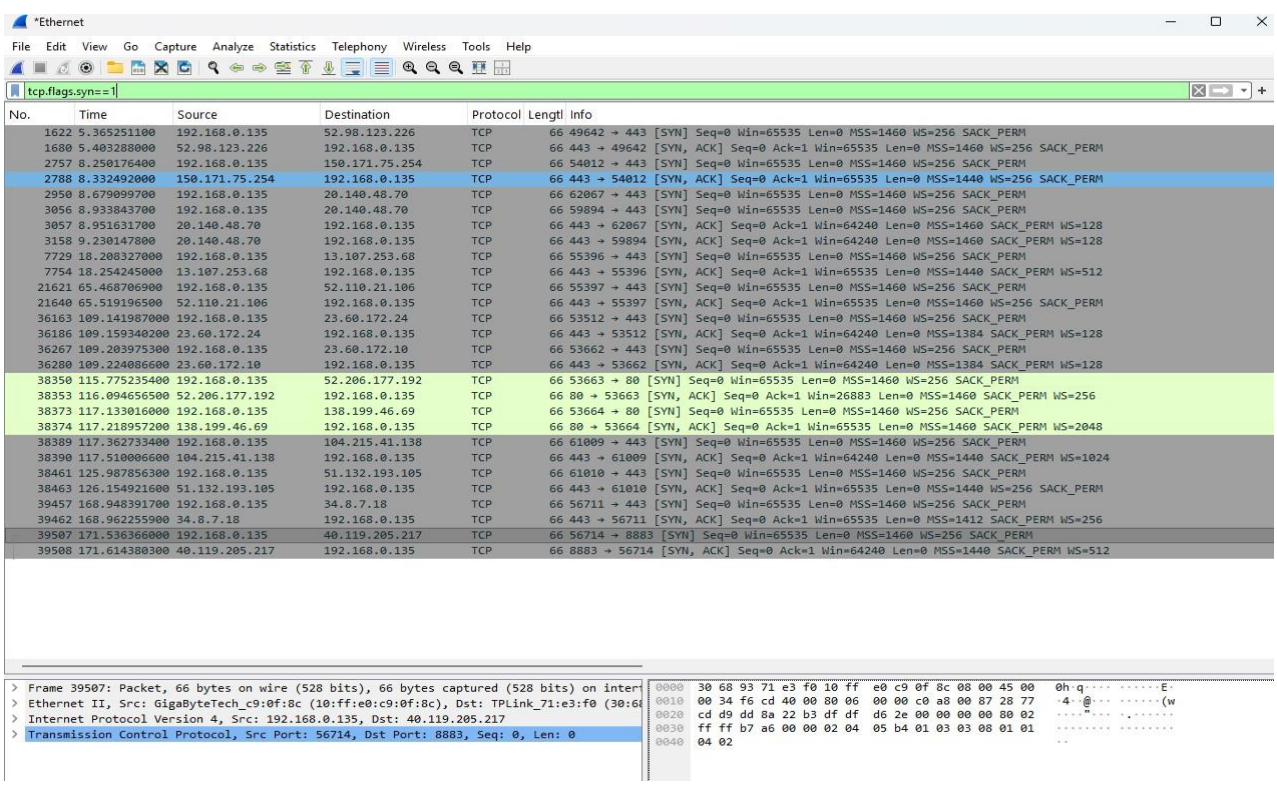
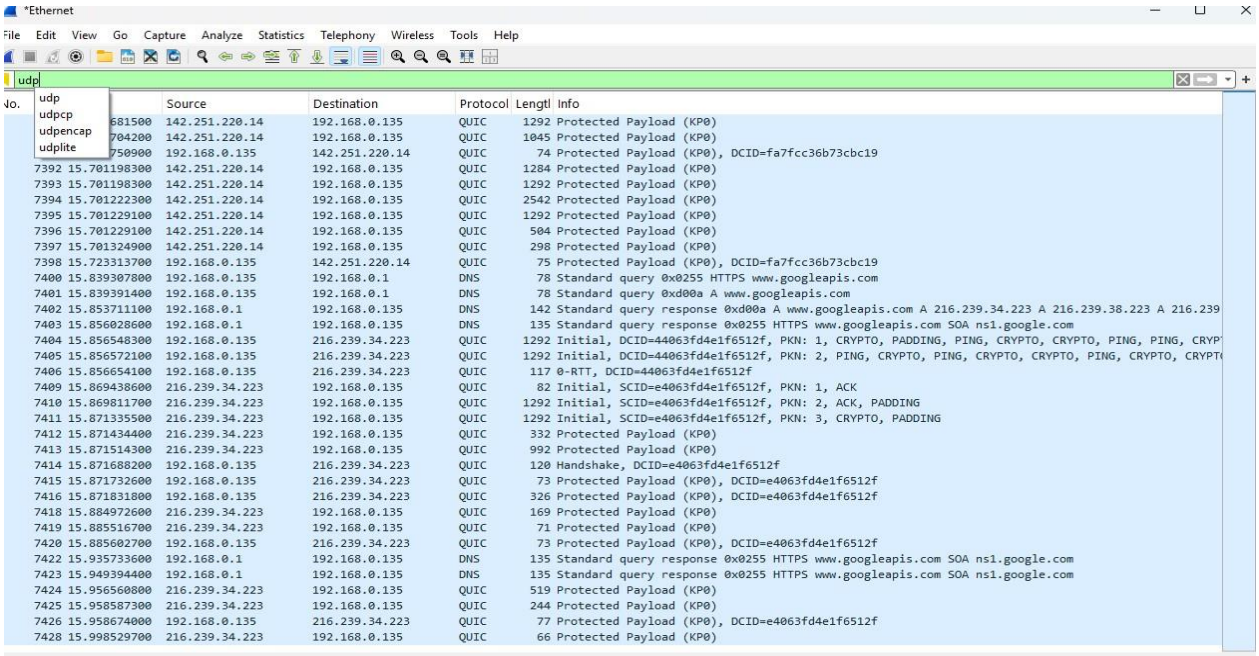
1. To capture and analyze ICMP packets (Ping request and reply)



2. To analyze TCP packets and observe the TCP three-way handshake

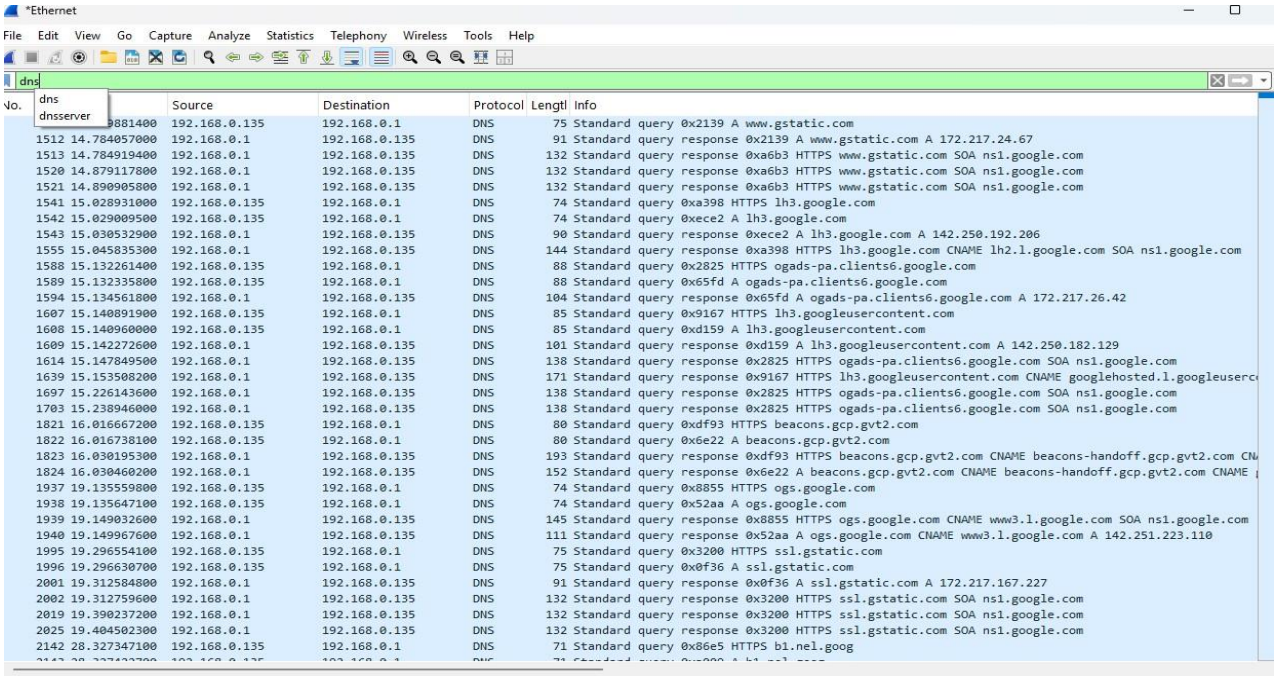


3. To capture and analyze UDP packets



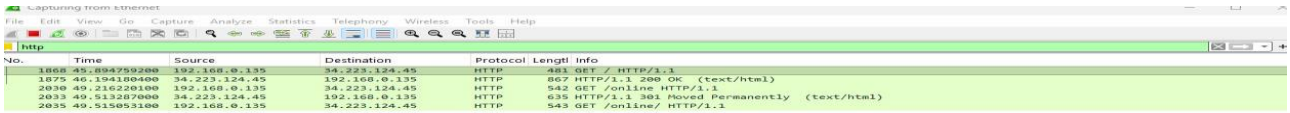
No.	Time	Source	Destination	Protocol	Length	Info
	681500	142.251.220.14	192.168.0.135	QUIC	1292	Protected Payload (KP0)
	704200	142.251.220.14	192.168.0.135	QUIC	1045	Protected Payload (KP0)
	750900	192.168.0.135	142.251.220.14	QUIC	74	Protected Payload (KP0), DCID=fa7fcc36b73cbc19
7392	15.701198300	142.251.220.14	192.168.0.135	QUIC	1284	Protected Payload (KP0)
7393	15.701198300	142.251.220.14	192.168.0.135	QUIC	1292	Protected Payload (KP0)
7394	15.701222300	142.251.220.14	192.168.0.135	QUIC	2542	Protected Payload (KP0)
7395	15.701229100	142.251.220.14	192.168.0.135	QUIC	1292	Protected Payload (KP0)
7396	15.701229100	142.251.220.14	192.168.0.135	QUIC	504	Protected Payload (KP0)
7397	15.701324900	142.251.220.14	192.168.0.135	QUIC	298	Protected Payload (KP0)
7398	15.723313700	192.168.0.135	142.251.220.14	QUIC	75	Protected Payload (KP0), DCID=fa7fcc36b73cbc19
7400	15.839307800	192.168.0.135	192.168.0.1	DNS	78	Standard query 0x0255 HTTPS www.googleapis.com
7401	15.839391400	192.168.0.135	192.168.0.1	DNS	78	Standard query 0xd00a A www.googleapis.com
7402	15.853711100	192.168.0.1	192.168.0.135	DNS	142	Standard query response 0xd00a A www.googleapis.com A 216.239.34.223 A 216.239.38.223 A 216.239
7403	15.856028600	192.168.0.1	192.168.0.135	DNS	135	Standard query response 0x0255 HTTPS www.googleapis.com SOA ns1.google.com
7404	15.856548300	192.168.0.135	216.239.34.223	QUIC	1292	Initial, DCID=44063fd4e1f6512f, PKN: 1, CRYPTO, PADDING, PING, CRYPTO, CRYPTO, PING, PING, CRYP
7405	15.856572100	192.168.0.135	216.239.34.223	QUIC	1292	Initial, DCID=44063fd4e1f6512f, PKN: 2, PING, CRYPTO, PING, CRYPTO, CRYPTO, PING, CRYPTO, CRYPT
7406	15.856654100	192.168.0.135	216.239.34.223	QUIC	117	0-RTT, DCID=44063fd4e1f6512f
7409	15.869438600	216.239.34.223	192.168.0.135	QUIC	82	Initial, SCID=e4063fd4e1f6512f, PKN: 1, ACK
7410	15.869811700	216.239.34.223	192.168.0.135	QUIC	1292	Initial, SCID=e4063fd4e1f6512f, PKN: 2, ACK, PADDING
7411	15.871335500	216.239.34.223	192.168.0.135	QUIC	1292	Initial, SCID=e4063fd4e1f6512f, PKN: 3, CRYPTO, PADDING
7412	15.871434400	216.239.34.223	192.168.0.135	QUIC	332	Protected Payload (KP0)
7413	15.871514300	216.239.34.223	192.168.0.135	QUIC	992	Protected Payload (KP0)
7414	15.871688200	192.168.0.135	216.239.34.223	QUIC	120	Handshake, DCID=e4063fd4e1f6512f
7415	15.871732600	192.168.0.135	216.239.34.223	QUIC	73	Protected Payload (KP0), DCID=e4063fd4e1f6512f
7416	15.871831800	192.168.0.135	216.239.34.223	QUIC	326	Protected Payload (KP0), DCID=e4063fd4e1f6512f
7418	15.884972600	216.239.34.223	192.168.0.135	QUIC	169	Protected Payload (KP0)
7419	15.885516700	216.239.34.223	192.168.0.135	QUIC	71	Protected Payload (KP0)
7420	15.885602700	192.168.0.135	216.239.34.223	QUIC	73	Protected Payload (KP0), DCID=e4063fd4e1f6512f
7422	15.935733600	192.168.0.1	192.168.0.135	DNS	135	Standard query response 0x0255 HTTPS www.googleapis.com SOA ns1.google.com
7423	15.949394400	192.168.0.1	192.168.0.135	DNS	135	Standard query response 0x0255 HTTPS www.googleapis.com SOA ns1.google.com
7424	15.956560800	216.239.34.223	192.168.0.135	QUIC	519	Protected Payload (KP0)
7425	15.958587300	216.239.34.223	192.168.0.135	QUIC	244	Protected Payload (KP0)
7426	15.958674000	192.168.0.135	216.239.34.223	QUIC	77	Protected Payload (KP0), DCID=e4063fd4e1f6512f
7428	15.998529700	216.239.34.223	192.168.0.135	QUIC	66	Protected Payload (KP0)

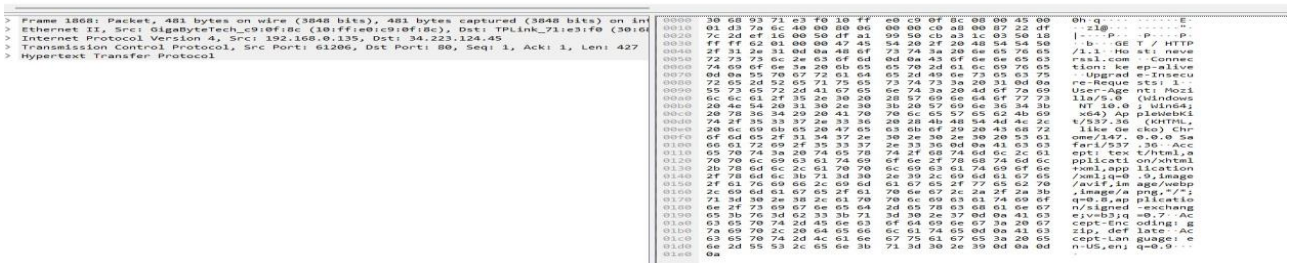
4. To capture and analyze DNS query and response packets



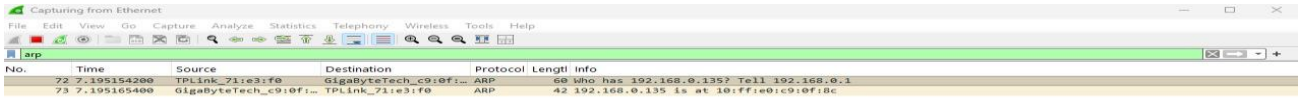
No.	Time	Source	Destination	Protocol	Length	Info
	881400	192.168.0.135	192.168.0.1	DNS	75	Standard query 0x2139 A www.gstatic.com
1512	14.784057000	192.168.0.1	192.168.0.135	DNS	91	Standard query response 0x2139 A www.gstatic.com A 172.217.24.67
1513	14.784919400	192.168.0.1	192.168.0.135	DNS	132	Standard query response 0xa6b3 HTTPS www.gstatic.com SOA ns1.google.com
1520	14.879117800	192.168.0.1	192.168.0.135	DNS	132	Standard query response 0xa6b3 HTTPS www.gstatic.com SOA ns1.google.com
1521	14.890905800	192.168.0.1	192.168.0.135	DNS	132	Standard query response 0xa6b3 HTTPS www.gstatic.com SOA ns1.google.com
1541	15.028931000	192.168.0.135	192.168.0.1	DNS	74	Standard query 0xa398 HTTPS lh3.google.com
1542	15.029009500	192.168.0.135	192.168.0.1	DNS	74	Standard query 0xece2 A lh3.google.com
1543	15.030532900	192.168.0.1	192.168.0.135	DNS	90	Standard query response 0xece2 A lh3.google.com A 142.250.192.206
1555	15.045835300	192.168.0.1	192.168.0.135	DNS	144	Standard query response 0xa398 HTTPS lh3.google.com CNAME lh2.l.google.com SOA ns1.google.com
1588	15.132261400	192.168.0.135	192.168.0.1	DNS	88	Standard query 0x2825 HTTPS ogads-pa.clients6.google.com
1589	15.132335800	192.168.0.135	192.168.0.1	DNS	88	Standard query 0x65fd A ogads-pa.clients6.google.com
1594	15.134561800	192.168.0.1	192.168.0.135	DNS	104	Standard query response 0x65fd A ogads-pa.clients6.google.com A 172.217.26.42
1607	15.140891900	192.168.0.135	192.168.0.1	DNS	85	Standard query 0x9167 HTTPS lh3.googleusercontent.com
1608	15.140960000	192.168.0.135	192.168.0.1	DNS	85	Standard query 0xd159 A lh3.googleusercontent.com
1609	15.142272600	192.168.0.1	192.168.0.135	DNS	101	Standard query response 0xd159 A lh3.googleusercontent.com A 142.250.182.129
1614	15.147849500	192.168.0.1	192.168.0.135	DNS	138	Standard query response 0x2825 HTTPS ogads-pa.clients6.google.com SOA ns1.google.com
1639	15.153508200	192.168.0.1	192.168.0.135	DNS	171	Standard query response 0x9167 HTTPS lh3.googleusercontent.com CNAME googlehosted.l.googleuserc
1697	15.226143600	192.168.0.1	192.168.0.135	DNS	138	Standard query response 0x2825 HTTPS ogads-pa.clients6.google.com SOA ns1.google.com
1703	15.238946000	192.168.0.1	192.168.0.135	DNS	138	Standard query response 0x2825 HTTPS ogads-pa.clients6.google.com SOA ns1.google.com
1821	16.016667200	192.168.0.135	192.168.0.1	DNS	80	Standard query 0xdf93 HTTPS beacons.gcp.gvt2.com
1822	16.016738100	192.168.0.135	192.168.0.1	DNS	80	Standard query 0x6e22 A beacons.gcp.gvt2.com
1823	16.030195300	192.168.0.1	192.168.0.135	DNS	193	Standard query response 0xdf93 HTTPS beacons.gcp.gvt2.com CNAME beacons-handoff.gcp.gvt2.com CN
1824	16.030460200	192.168.0.1	192.168.0.135	DNS	152	Standard query response 0x6e22 A beacons.gcp.gvt2.com CNAME beacons-handoff.gcp.gvt2.com CNAME
1937	19.135559800	192.168.0.135	192.168.0.1	DNS	74	Standard query 0x8855 HTTPS ogs.google.com
1938	19.135647100	192.168.0.135	192.168.0.1	DNS	74	Standard query 0x52aa A ogs.google.com
1939	19.149032600	192.168.0.1	192.168.0.135	DNS	145	Standard query response 0x8855 HTTPS ogs.google.com CNAME www3.l.google.com SOA ns1.google.com
1940	19.149967600	192.168.0.1	192.168.0.135	DNS	111	Standard query response 0x52aa A ogs.google.com CNAME www3.l.google.com A 142.251.223.110
1995	19.296554100	192.168.0.135	192.168.0.1	DNS	75	Standard query 0x3200 HTTPS ssl.gstatic.com
1996	19.296630700	192.168.0.135	192.168.0.1	DNS	75	Standard query 0x0f36 A ssl.gstatic.com
2001	19.312584800	192.168.0.1	192.168.0.135	DNS	91	Standard query response 0x0f36 A ssl.gstatic.com A 172.217.167.227
2002	19.312759600	192.168.0.1	192.168.0.135	DNS	132	Standard query response 0x3200 HTTPS ssl.gstatic.com SOA ns1.google.com
2019	19.390237200	192.168.0.1	192.168.0.135	DNS	132	Standard query response 0x3200 HTTPS ssl.gstatic.com SOA ns1.google.com
2025	19.404502300	192.168.0.1	192.168.0.135	DNS	132	Standard query response 0x3200 HTTPS ssl.gstatic.com SOA ns1.google.com
2142	28.327347100	192.168.0.135	192.168.0.1	DNS	71	Standard query 0x86e5 HTTPS b1.nel.goog

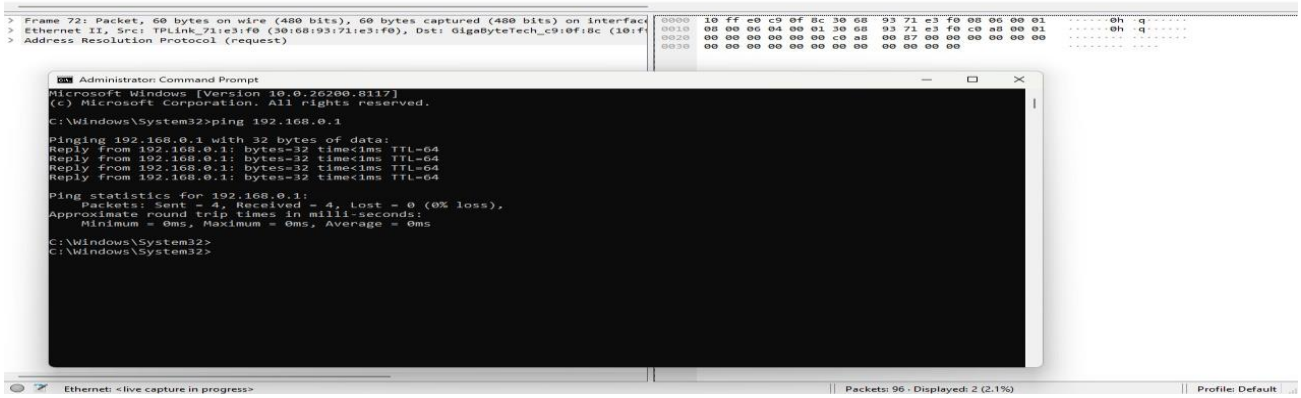
5. To capture and analyze HTTP packets





6. To analyze ARP request and reply packets





*Ethernet

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Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Lengtl	Info
1142	59.466297600	23.60.172.24	192.168.0.135	TLSv1.2	78	Applicati
1143	59.466461000	192.168.0.135	23.60.172.24	TCP	54	61945 → 4
1144	59.466614100	23.60.172.24	192.168.0.135	TCP	60	443 → 619
1145	59.466630500	192.168.0.135	23.60.172.24	TCP	54	61945 → 4
1146	59.484870000	23.60.172.24	192.168.0.135	TCP	60	443 → 619
1147	59.826002800	192.168.0.135	142.250.122.94	TCP	55	[TCP Keep
1148	59.840840700	142.250.122.94	192.168.0.135	TCP	66	[TCP Keep
1149	59.966791800	23.60.172.24	192.168.0.135	TLSv1.2	78	Applicati
1150	59.966928900	192.168.0.135	23.60.172.24	TCP	54	62305 → 4
1151	59.967163200	23.60.172.24	192.168.0.135	TCP	60	443 → 623
1152	59.967193300	192.168.0.135	23.60.172.24	TCP	54	62305 → 4
1153	59.985525300	23.60.172.24	192.168.0.135	TCP	60	443 → 623
1154	60.007822500	104.18.36.252	192.168.0.135	TLSv1.2	188	Applicati
1155	60.007973800	192.168.0.135	104.18.36.252	TLSv1.2	82	Applicati
1156	60.008195600	192.168.0.135	104.18.36.252	TLSv1.2	174	Applicati
1157	60.019878500	104.18.36.252	192.168.0.135	TCP	60	443 → 637
1158	60.019878500	104.18.36.252	192.168.0.135	TCP	60	443 → 637
1159	60.170488200	192.168.0.135	35.186.224.39	TLSv1.2	97	Applicati
1160	60.184859200	35.186.224.39	192.168.0.135	TCP	60	443 → 555
1161	60.293266400	35.186.224.39	192.168.0.135	TLSv1.2	94	Applicati
1162	60.341099200	192.168.0.135	35.186.224.39	TCP	54	55515 → 4
1163	60.482539400	192.168.0.135	192.168.0.255	UDP	86	57621 → 5
1164	60.754787400	216.58.196.106	192.168.0.135	UDP	200	443 → 526
1165	60.781914300	192.168.0.135	216.58.196.106	UDP	74	52659 → 4
1166	62.899374600	192.168.0.1	239.255.255.250	SSDP	448	NOTIFY *
1167	62.993405500	192.168.0.135	162.159.133.233	TCP	55	[TCP Keep
1168	63.003524900	162.159.133.233	192.168.0.135	TCP	66	[TCP Keep
1169	63.009450800	192.168.0.1	239.255.255.250	SSDP	520	NOTIFY *
1170	63.009619300	192.168.0.1	239.255.255.250	SSDP	457	NOTIFY *
1171	63.119350800	192.168.0.1	239.255.255.250	SSDP	516	NOTIFY *
1172	63.119502400	192.168.0.1	239.255.255.250	SSDP	457	NOTIFY *
1173	63.229444000	192.168.0.1	239.255.255.250	SSDP	496	NOTIFY *
1174	63.229600300	192.168.0.1	239.255.255.250	SSDP	457	NOTIFY *
1175	63.339431100	192.168.0.1	239.255.255.250	SSDP	510	NOTIFY *

```
> Frame 1150: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on inter
> Ethernet II, Src: GigaByteTech_c9:0f:8c (10:ff:e0:c9:0f:8c), Dst: TPLink_71:e3:f0 (30:
> Internet Protocol Version 4, Src: 192.168.0.135, Dst: 23.60.172.24
> Transmission Control Protocol, Src Port: 62305, Dst Port: 443, Seq: 2, Ack: 25, Len: 0
```